

Table 3
Mean absolute error (MAE) of Ishii–Zuber viscous drag correlation (Ishii and Zuber, 1979) with respect to experimental bubble terminal Reynolds numbers.

Test case (W/cm ²)	Number of points	MAE (%) ^a	Notes
Polished, 2	26	14.5	Slightly underpredicted Re _{b,term}
Polished, 5	25	36.5	Overpredicted Re _{b,term}
Polished, 8	25	45.3	Overpredicted Re _{b,term}
Polished, 11	24	47.8	Overpredicted Re _{b,term} ; error not Gaussian
EDM, 2	26	30.0	Underpredicted Re _{b,term}
EDM, 5	25	27.3	Slightly underpredicted Re _{b,term}
EDM, 8	21	29.7	Slightly overpredicted Re _{b,term}
EDM, 11	12	25.2	Underpredicted Re _{b,term} ; frequent coalescences obscured velocity meas

^a MAE is defined as the sum of the absolute deviations of the correlation with respect to the experimentally measured values, divided by the sum of the experimentally measured values.

velocity is based upon the measured bubble diameter at departure according to:

$$v_f = \dot{y}_f = \sqrt{\left(\frac{4}{3}\right) \frac{gD_d}{C_D} \left(\frac{\rho_l - \rho_v}{\rho_l}\right)}. \tag{5}$$

The theoretical drag coefficients may be evaluated with reasonable agreement according to the viscous drag correlation (Ishii and Zuber, 1979):

$$C_D = \frac{24}{Re} (1 + 0.1Re^{0.75}), \tag{6}$$

where the Reynolds number is the mixture-viscosity corrected value, with mixture viscosity μ_{mix} obtained by the expression:

$$\frac{\mu_{mix}}{\mu_l} = \left(1 - \frac{\alpha_v}{\alpha_{v,max}}\right)^{-2.5\alpha_{v,max}(\mu_v + 0.4\mu_l)/(\mu_v + \mu_l)}. \tag{7}$$

A number of additional terms are introduced in Eq. (7) and require further explanation. Void fraction, α_v , may attain a maximum value, $\alpha_{v,max}$, of 1. The void fraction was not measured directly, but was estimated to first order as the ratio of the evaporative component of heat flux to the evaporative component of the maximum heat flux. The evaporative component itself was estimated as the total heat flux minus the natural convection heat flux predicted for each wall superheat temperature. The resulting void fraction values were consistent with those measured for water at low, medium and high heat fluxes by Iida and Kobayasi (1969). The vapor viscosity for FC-77 is not available in the literature, but it is a reasonable assumption for most gases at atmospheric pressure to use the estimate $\mu_v \approx 2 \times 10^{-5}$ kg/m s.

In Table 3, the mean absolute error (MAE) of the Ishii–Zuber (1979) correlation with respect to the experimental data is given along with the number of experimental values included in the comparison. The observed bubble mixture Reynolds numbers ranged between 15 and 1250. The good agreement (less than 50% MAE) shows that the assumption of a quiescent pool is reasonable and that the frequent bubble interactions do not prevent the conventional two-phase drag model from being used in an averaged sense, even for rather high heat fluxes. It is worth noting that slightly better agreement could be obtained for high heat fluxes using the Newton’s regime drag model proposed in (Ishii and Zuber, 1979). This model, however, is not applicable for smaller bubbles.

6. Conclusions

Important bubble nucleation parameters during the pool boiling of FC-77 on smooth and rough surfaces are experimentally investigated. The pool boiling observed was characterized by high bubble densities and numerous interactions between bubbles. The characteristics of bubble nucleation from the two surfaces at vari-

ous heat fluxes were compared qualitatively. Measurements were obtained from high-speed movies for 25 individual bubbles per test case, comprised of five consecutive bubbles from each of five nucleation sites on the surface.

Bubble diameter at departure was shown to increase with increasing wall superheat, but the surface roughness was also shown to have an influence. Bubble departure frequency, which in general also increased with heat flux, was not well predicted by any of the correlations considered from the literature. Active nucleation site density was shown to increase with both wall superheat and surface roughness. Active site density was reasonably well predicted for both the polished and roughened surfaces by the Hibiki–Ishii (2003) correlation. Terminal rise velocity increased with increasing heat flux due to the increased buoyancy from larger bubble volumes; average pool boiling bubble velocities were shown to be well predicted by the Ishii–Zuber (1979) mixture-viscosity based correlation for viscous drag.

It is suggested that new bubble nucleation correlations be developed which incorporate the important effect of surface roughness, so that recent developments in mechanistic modeling can be applied for a broad range of boiling surfaces.

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References

3M Corporation, 1986. Fluorinert liquids product manual. 3M Corporation.
3M Corporation, 2000. Fluorinert electronic liquid FC-77 product information. 3M Corporation, Document 98-0212-2309-8 (HB).
Abarajith, H.S., Dhir, V.K., Warriier, G., Son, G., 2004. Numerical simulation and experimental validation of the dynamics of multiple bubble merger during pool boiling under microgravity conditions. *Transport Phenomena in Microgravity* 1027, 235–258.
Benjamin, R., Balakrishnan, A., 1996. Nucleate pool boiling heat transfer of pure liquids at low to moderate heat fluxes. *International Journal of Heat and Mass Transfer* 39, 2495–2504.
Bernardin, J.D., Mudawar, I., Walsh, C.B., Franes, E.I., 1997. Contact angle temperature dependence for water droplets on practical aluminum surfaces. *International Journal of Heat and Mass Transfer* 40, 1017–1033.
Bier, K., Gorenflo, D., Salem, M., Tanes, Y., 1979. Effect of pressure and surface roughness on pool boiling of refrigerants. *International Journal of Refrigeration* 2, 211–219.
Borishanskiy, V., Danilova, G., Gotovskiy, M., Borishanskiy, A., Danilova, G., Kupriyanova, A., 1981. Correlation of data on heat transfer in and elementary characteristics of the nucleate boiling mechanism. *Heat Transfer – Soviet Research* 13, 100–116.
Chekanov, V., 1977. Interaction of centers during nucleate boiling. *Teplofizika Vysokikh Temperature* 15, 121–128.